

In []:

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline
```

```
data=pd.read_csv('/content/Iris_Dataset.csv')
data
```

```
↵
```

|     | Id  | SepalLengthCm | SepalWidthCm | PetalLengthCm | PetalWidthCm | variety        |
|-----|-----|---------------|--------------|---------------|--------------|----------------|
| 0   | 1   | 5.1           | 3.5          | 1.4           | 0.2          | Iris-setosa    |
| 1   | 2   | 4.9           | 3.0          | 1.4           | 0.2          | Iris-setosa    |
| 2   | 3   | 4.7           | 3.2          | 1.3           | 0.2          | Iris-setosa    |
| 3   | 4   | 4.6           | 3.1          | 1.5           | 0.2          | Iris-setosa    |
| 4   | 5   | 5.0           | 3.6          | 1.4           | 0.2          | Iris-setosa    |
| ... | ... | ...           | ...          | ...           | ...          | ...            |
| 145 | 146 | 6.7           | 3.0          | 5.2           | 2.3          | Iris-virginica |
| 146 | 147 | 6.3           | 2.5          | 5.0           | 1.9          | Iris-virginica |
| 147 | 148 | 6.5           | 3.0          | 5.2           | 2.0          | Iris-virginica |
| 148 | 149 | 6.2           | 3.4          | 5.4           | 2.3          | Iris-virginica |
| 149 | 150 | 5.9           | 3.0          | 5.1           | 1.8          | Iris-virginica |

150 rows x 6 columns

```
data.info()
```

```
↵
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
#   Column             Non-Null Count  Dtype
---  -
0   Id                  150 non-null   int64
1   SepalLengthCm       150 non-null   float64
2   SepalWidthCm        150 non-null   float64
3   PetalLengthCm       150 non-null   float64
4   PetalWidthCm        150 non-null   float64
5   variety             150 non-null   object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
```

```
data.describe()
```

```
↵
```

|       | Id         | SepalLengthCm | SepalWidthCm | PetalLengthCm | PetalWidthCm |
|-------|------------|---------------|--------------|---------------|--------------|
| count | 150.000000 | 150.000000    | 150.000000   | 150.000000    | 150.000000   |
| mean  | 75.500000  | 5.843333      | 3.054000     | 3.758667      | 1.198667     |
| std   | 43.445368  | 0.828066      | 0.433594     | 1.764420      | 0.763161     |
| min   | 1.000000   | 4.300000      | 2.000000     | 1.000000      | 0.100000     |
| 25%   | 38.250000  | 5.100000      | 2.800000     | 1.600000      | 0.300000     |
| 50%   | 75.500000  | 5.800000      | 3.000000     | 4.350000      | 1.300000     |
| 75%   | 112.750000 | 6.400000      | 3.300000     | 5.100000      | 1.800000     |
| max   | 150.000000 | 7.900000      | 4.400000     | 6.900000      | 2.500000     |

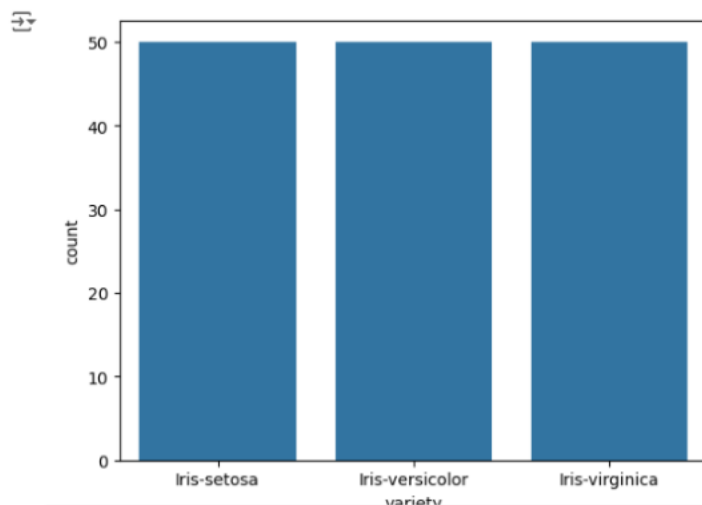
```
data.value_counts('variety')
```

```
↵
```

|                 | count |
|-----------------|-------|
| variety         |       |
| Iris-setosa     | 50    |
| Iris-versicolor | 50    |
| Iris-virginica  | 50    |

In[]:

```
sns.countplot(x='variety',data=data,)
plt.show()
```



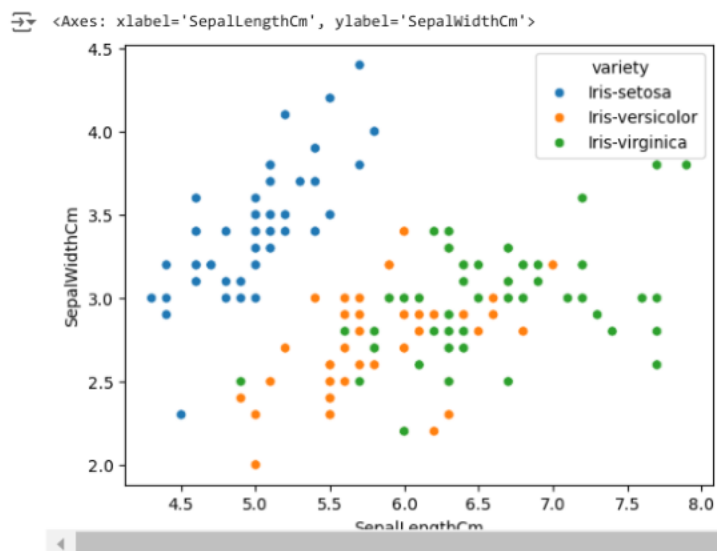
```
dummies=pd.get_dummies(data.variety)
```

```
FinalDataset=pd.concat([pd.get_dummies(data.variety),data.iloc[:,[0,1,2,3]]],axis=1)
```

```
FinalDataset.head()
```

|   | Iris-setosa | Iris-versicolor | Iris-virginica | Id | SepalLengthCm | SepalWidthCm | PetalLengthCm |
|---|-------------|-----------------|----------------|----|---------------|--------------|---------------|
| 0 | True        | False           | False          | 1  | 5.1           | 3.5          | 1.4           |
| 1 | True        | False           | False          | 2  | 4.9           | 3.0          | 1.4           |
| 2 | True        | False           | False          | 3  | 4.7           | 3.2          | 1.3           |
| 3 | True        | False           | False          | 4  | 4.6           | 3.1          | 1.5           |
| 4 | True        | False           | False          | 5  | 5.0           | 3.6          | 1.4           |

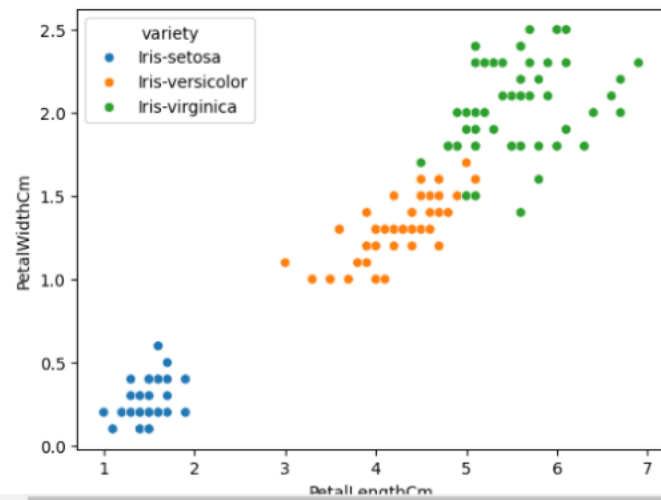
```
sns.scatterplot(x='SepalLengthCm',y='SepalWidthCm',hue='variety',data=data,)
```



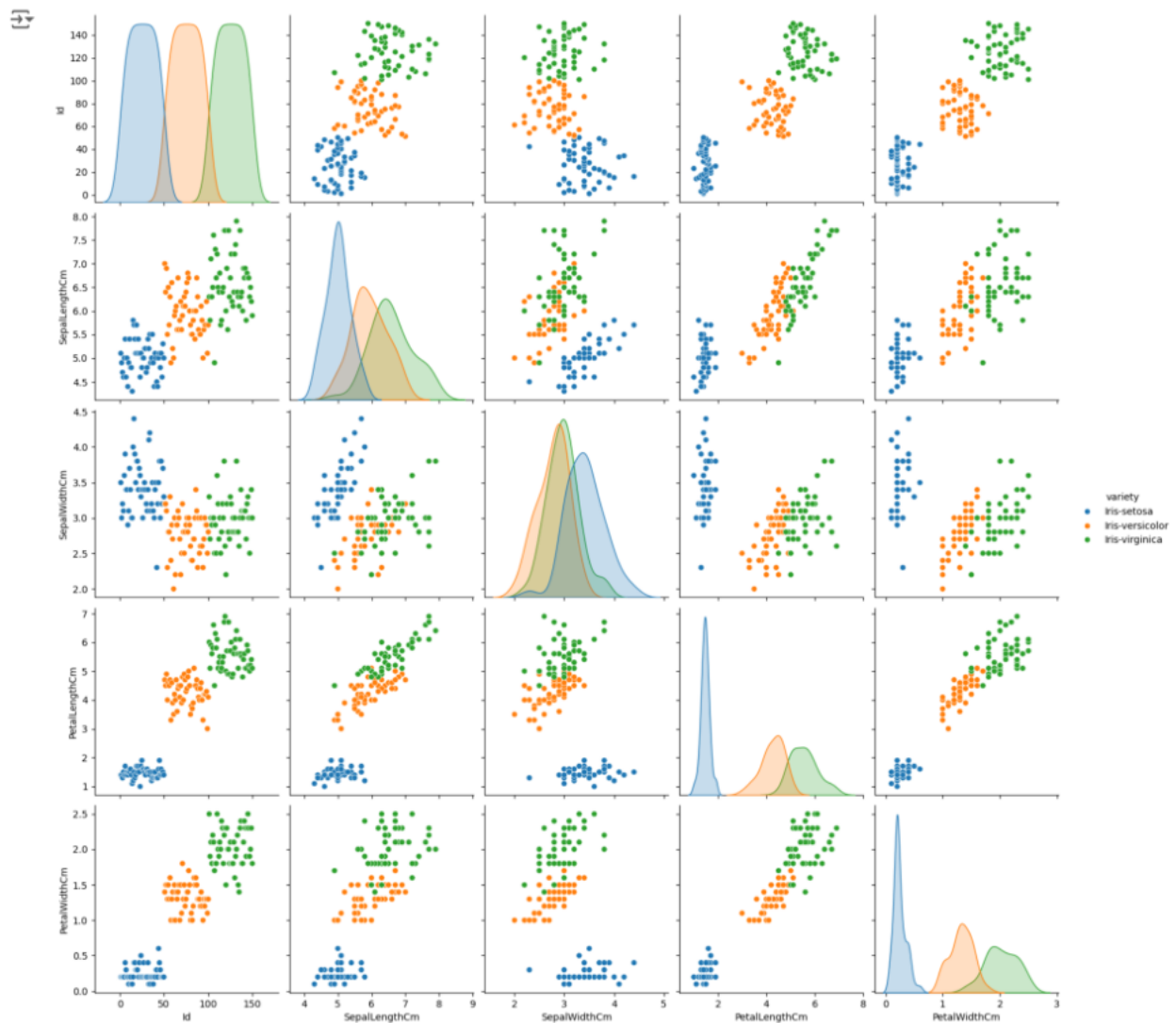
```
sns.scatterplot(x='PetalLengthCm',y='PetalWidthCm',hue='variety',data=data,)
```

In[]:

```
<Axes: xlabel='PetalLengthCm', ylabel='PetalWidthCm'>
```



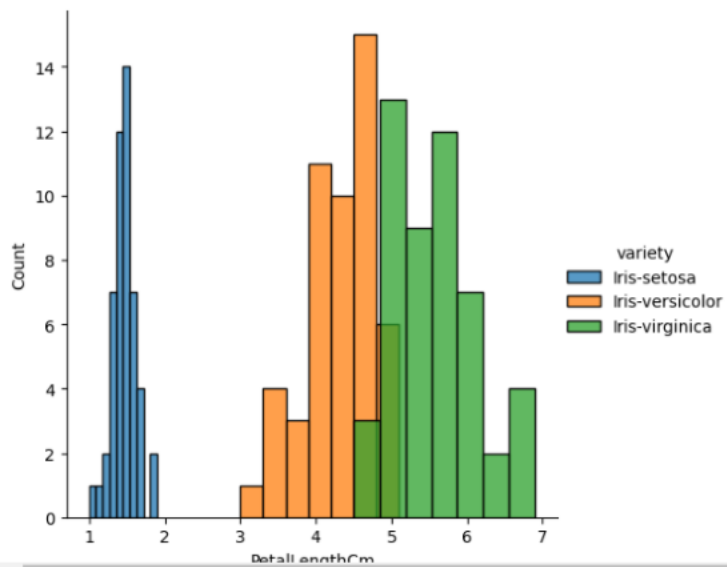
```
sns.pairplot(data,hue='variety',height=3);
```



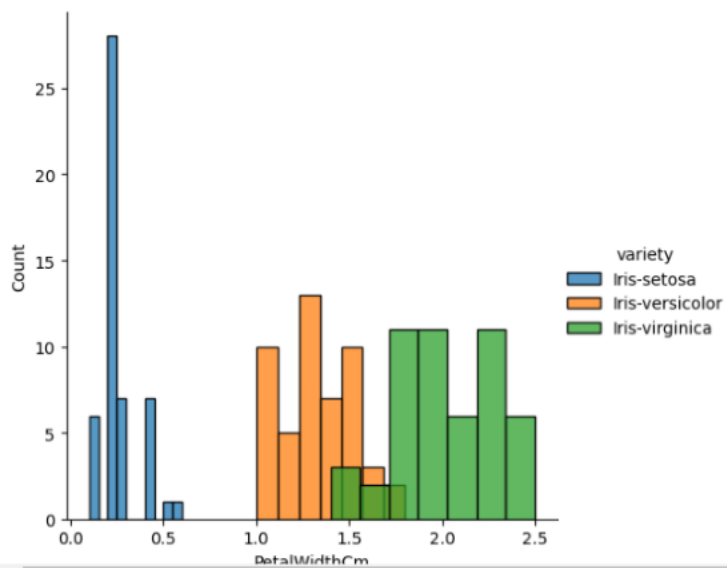
In[]

```
plt.show()
```

```
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'PetalLengthCm').add_legend();  
plt.show();
```

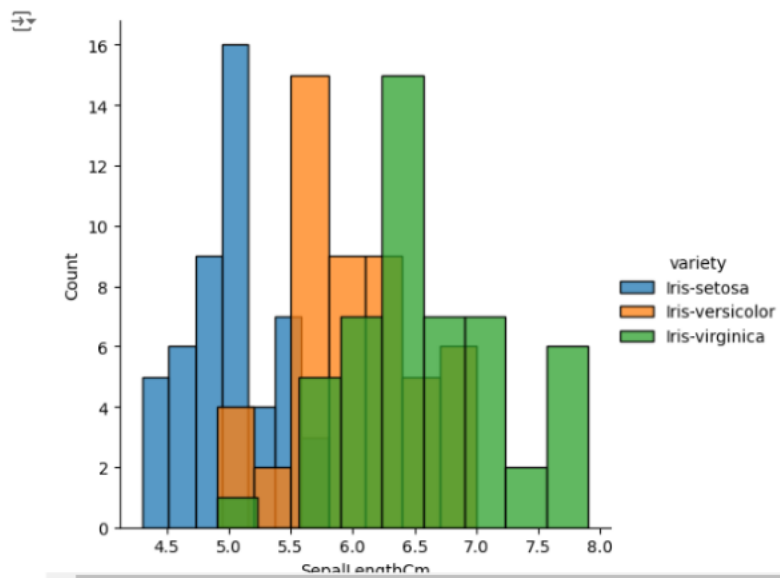


```
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'PetalWidthCm').add_legend();  
plt.show();
```



```
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'SepalLengthCm').add_legend();  
plt.show();
```

In []:



```
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'SepalWidthCm').add_legend();  
plt.show();
```

